Composable Coresets for Fair Diversity Maximization

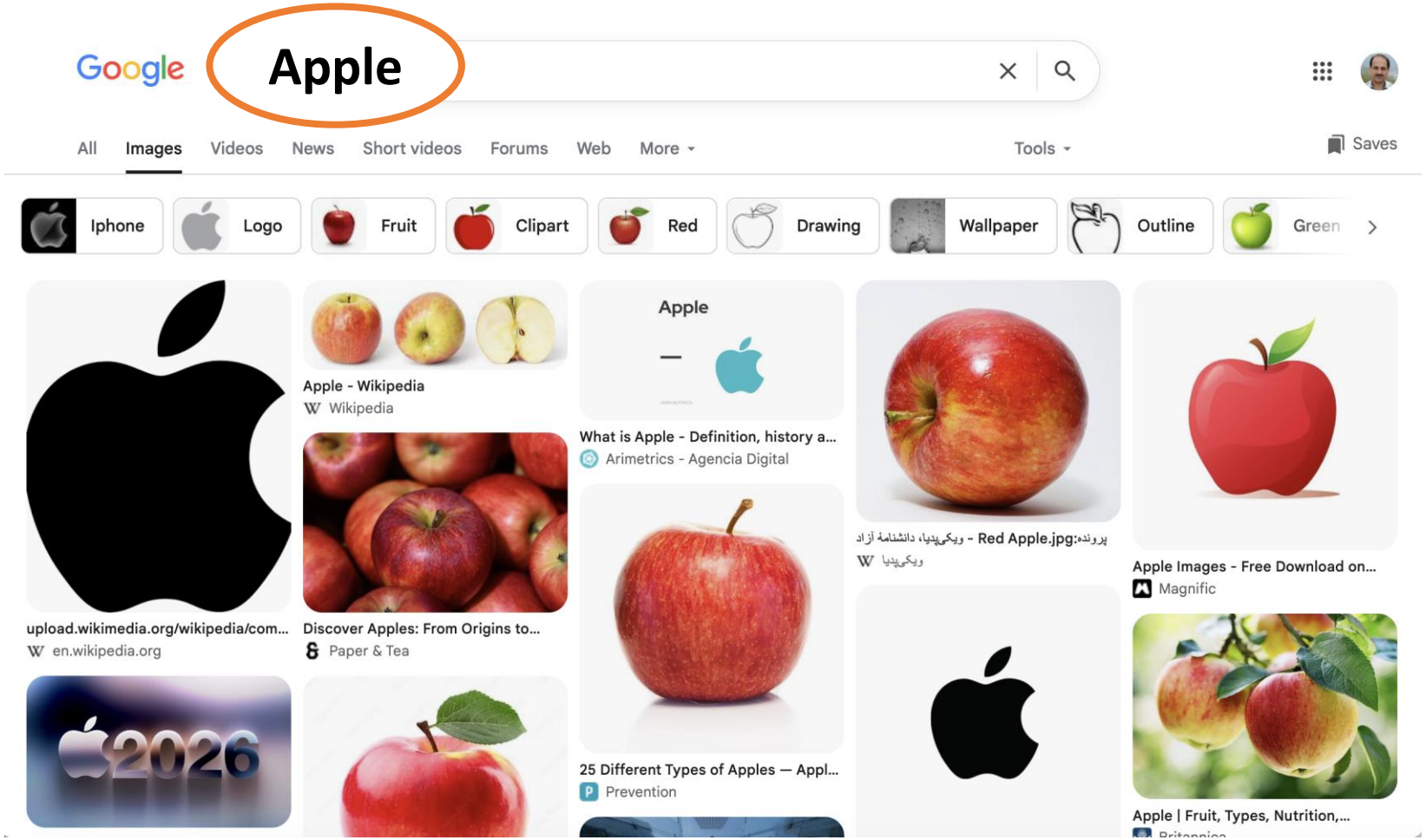
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SPAA 2026

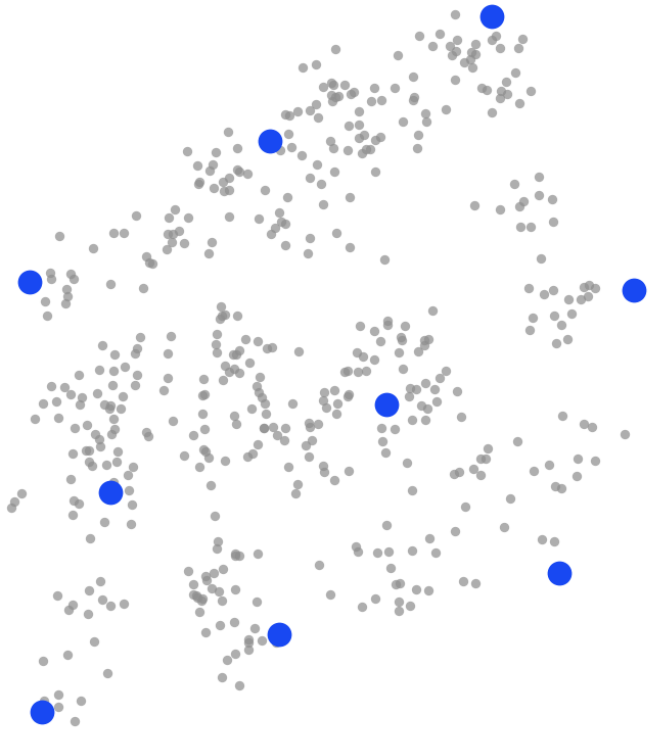
Diversity Maximization

- Search results

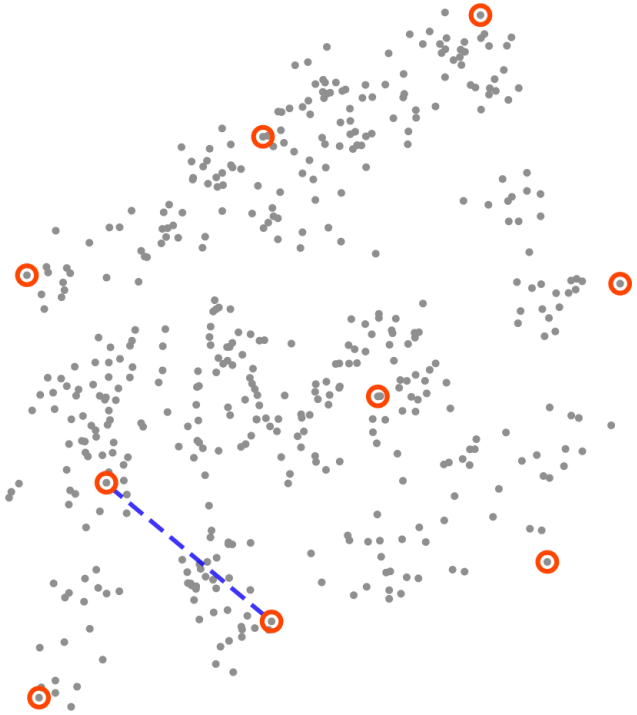


Diversity Maximization: Definition

Given a set P of n points in a metric space return a subset of k points maximizing a defined notion of diversity.

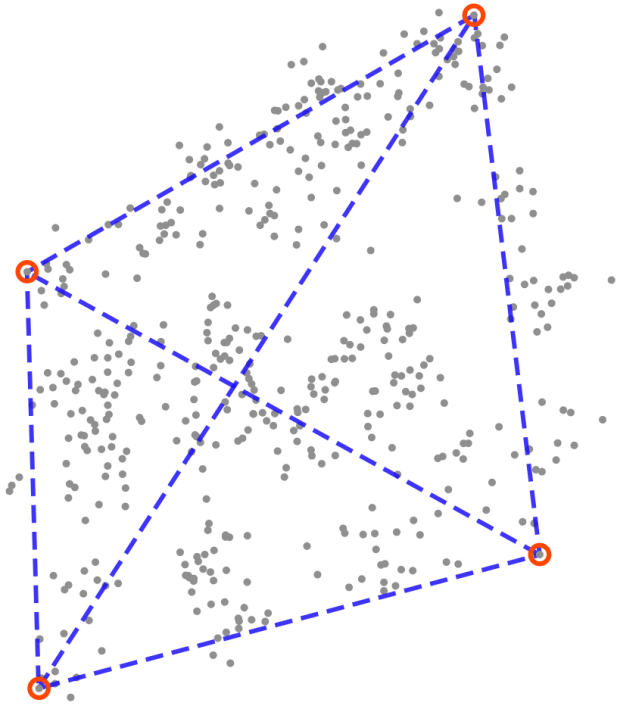


Diversity Notions



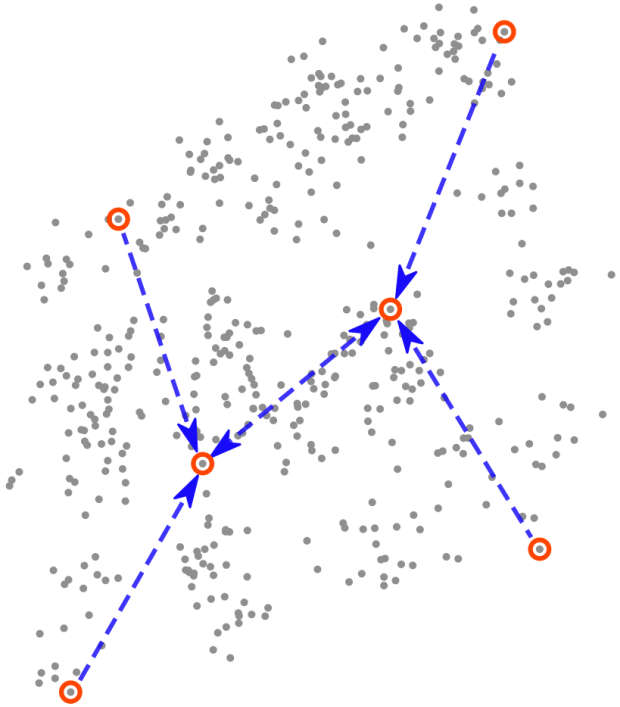
Max-Min diversity

for $k = 9$



Max-Sum diversity

for $k = 4$



Sum-NN diversity

for $k = 6$

Diversity Notions (cont'd)

- Max-Min: minimum pairwise distance of the points
- Max-Sum: sum of pairwise distances of the points
- Sum-NN: sum of distances from each point to its nearest neighbor

All three versions are NP-Hard.

Applications

- Recommendation systems
  - Shopping, movies, …
- Document and data summarization
- Search results
- Facility placement
- Machine learning dataset selection



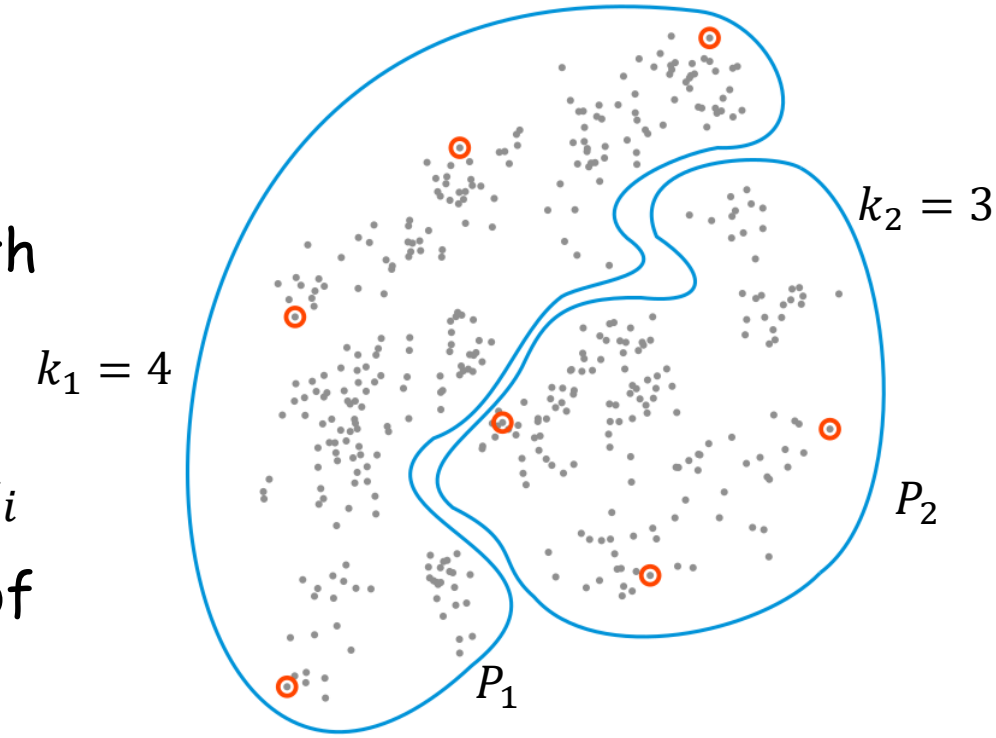
Image from: https://www.marketresearchintellect.com/images/blogs/driving-personalized-journeys-how-content-recommendation-engines-are-revolutionizing-the-auto-industry.webp

Fair Diversity Maximization

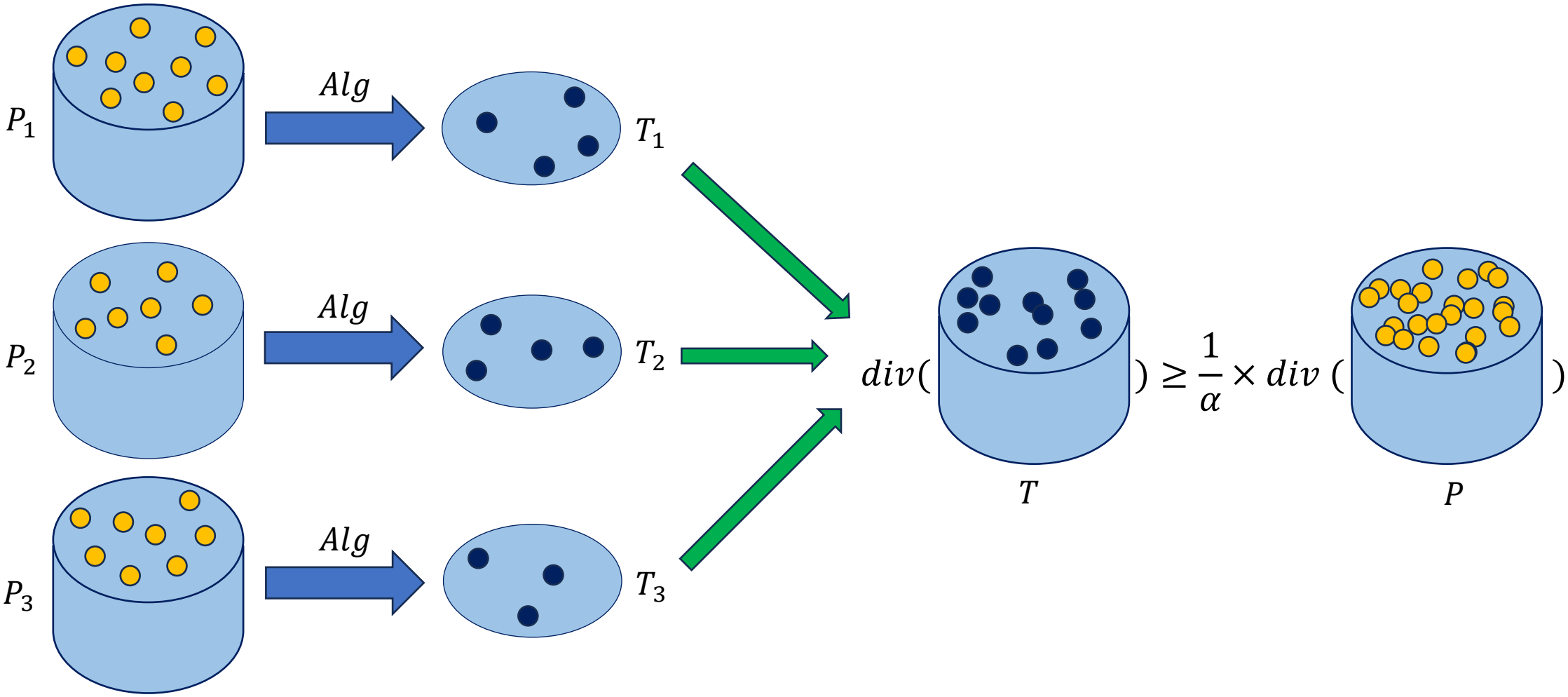
Given a set P of n points in a metric space, partitioned across m machines.

Let P_i be the set of points assigned to the i-th machine.

Find a subset of k points containing exactly k_i points from P_i, maximizing a defined notion of diversity.



Composable Coresets



Previous Work: Diversity Maximization

Notion of diversity	Approx factor (offline setting)	Reference	Coreset		
			Approx.	Size	Ref
Max-Min	$O(1)$	[Gon'85, RRT'94]	$O(1)$	k	[IMMM'14]
Max-Sum	$O(1)$	[HRT'97]	$O(1)$	k	[IMMM'14]
Sum-NN	$O(1)$	[BGMS'16]	$O(\frac{1}{\epsilon})$	$O(k^{1+\epsilon})$	[MN'23]

Previous Work: Fair Diversity Maximization

Notion of diversity	Approx factor (offline setting)	Reference	Coreset		
			Approx.	Size	Ref
Max-Min	$O(m)$	[MMM20, AMMM'22]	$O(1)$	$O(k)$	[MMM'20]
Max-Sum	$O(1)$	[AMM'13]	$O(1)$	$O(k_i^2)$	[MT'23]
Sum-NN	$O(1)$	[BGMS'16]	$O(m \log k)$	$O(k^2)$	[MT'23]

Our Contribution

Better Composable Coresets for Fair Diversity Maximization

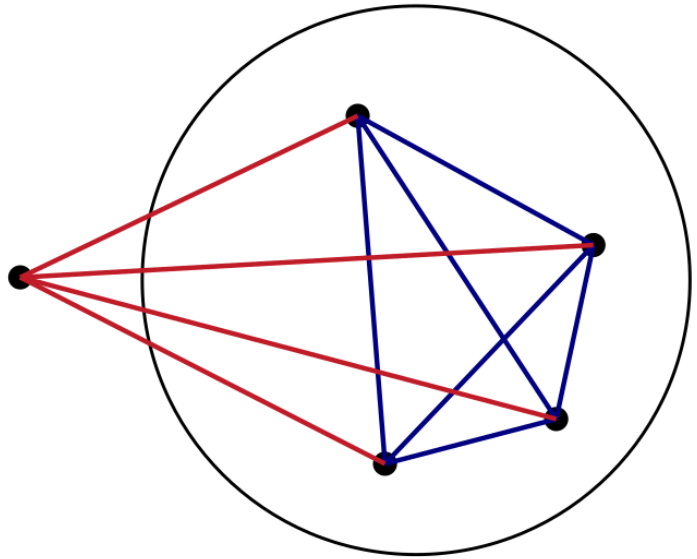
Measure	Previous			New		
	Approx.	Size per machine	Total size	Approx.	Size per machine	Total size
Max-Sum	320	$O(k_i^2)$	$O(\sum k_i^2)$	$5+\epsilon$	$O(k_i)$	$O(k)$
Sum-NN	$O(m \log k)$	$O(k^2)$	$O(mk^2)$	10	$O(kk_i)$	$O(k^2)$

Coreset for Max-Sum

- An improved $5 + \epsilon$-composable coreset
- Time complexity linear in terms of $\frac{1}{\epsilon}$
- Reduced local coreset size from quadratic to linear in k_i
- Returning a solution rather than a coreset if $k_i \geq 2$ for all machines

Coreset for Max-Sum (cont'd)

- Bounding sum of edges between points assigned to different machines
  - Δ-inequality for any point out of a machine and any pair of points within it
- Maximizing sum of edges for points selected by each machine
  - Local search approach



Coreset for Max-Sum (cont'd)

Challenge: machines required to have exactly one point in the solution

- No edges in this machine to use for bounding

Return the furthest pair of points from the ones assigned to it

- Calculating their contribution to diversity for each choice
- Using pigeonhole principle to show a proper choice exists

A solution is constructed if $k_i \geq 2$ for all machines.

Coreset for Sum-NN

- A new 10-composable coreset
- First constant-factor algorithm, improving from $O(m \log k)$
- Total coreset size at most k^2

Coreset Construction

Coreset selection on each P_i:

- Apply GMM to find k centers
- Add k_i points from the cluster around each center (if possible)
- Include the center itself in those points

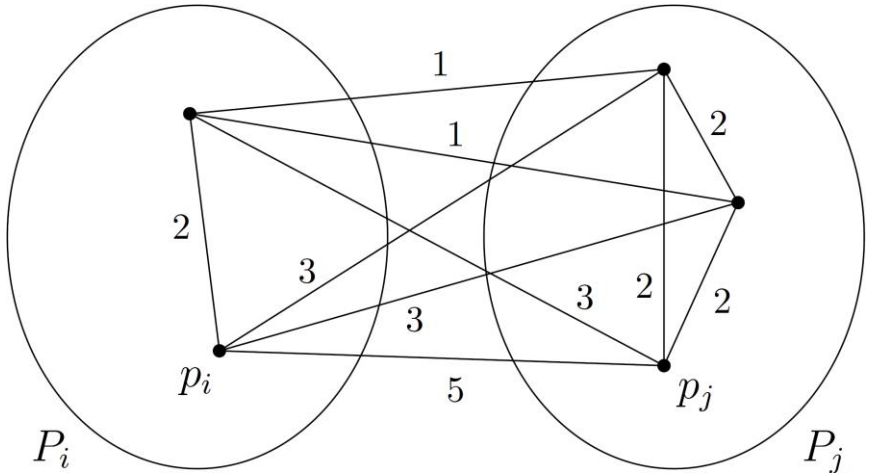
Proof Sketch

- Extract a solution Q_1
  - Constructing iteratively from the GMM centers
  - Diversity bounded in terms of cluster diameters
- Take an arbitrary optimal solution Q^*
- Extract another solution Q_2
  - Mapping from Q^* to coreset points
  - Points being mapped to the same clusters
  - Each mapped distance $\leq$ cluster diameter

- Combining two relations
  - A 10-approx solution found.

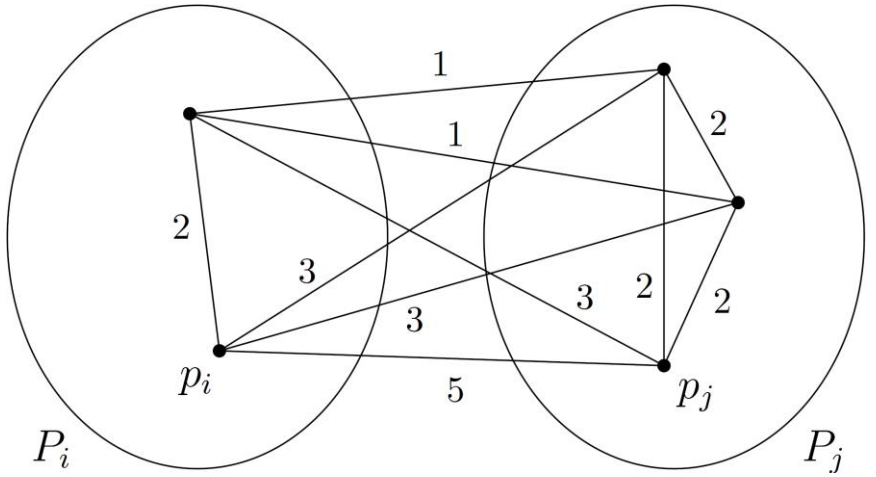
Lower Bound

- Target measures:
  - Sum of a specific number of pairwise distances
  - Max-Min, Max-Sum, Sum-NN, …
- Construction:
  - Any two points assigned to one machine have same distance
  - No difference between points



Lower Bound (cont'd)

- One point is designated in each machine.
  - Any two designated points ⇒ 5
  - Any two non-designated points ⇒ 1
- Impossible to guarantee to return the designated ones
- Lower bound of 5 for sublinear coresets



Summary

- Improved coresets for Fair Diversity Maximization
  - Max-Sum
    - Improved approx. factor with smaller coreset size
    - Almost optimal approx. factor
    - Returns a solution if $k_i \geq 2$ for all machines
  - Sum-NN
    - Improved approx. factor with smaller coreset size
    - First constant-factor coreset
- A universal lower bound for a variety of measures

Open Problems

- Can the approximation factor of either measure be improved to the optimal bound proved here?
- Is it possible to reduce the coreset size to $O(k)$ for Sum-NN measure while preserving constant approximation factor?
- Can techniques developed in this work be applied to other measures?

Thank you!